The Chain Rule & Directional Derivatives

ANSWER KEY

The multivariable chain rule

1 Let $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$.
$\frac{\partial z}{\partial x} = 2x$, $\frac{\partial z}{\partial y} = 2y$; $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$. $\frac{dz}{dt} = 2x(-\sin t) + 2y\cos t = -2\cos t\sin t + 2\sin t\cos t = 0$ (consistent with $z = \cos^2 t + \sin^2 t = 1$, a constant).

2 Let $w = xy + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
$\frac{\partial w}{\partial x} = y$, $\frac{\partial w}{\partial y} = x$, $\frac{\partial w}{\partial z} = 2z$. $\frac{dw}{dt} = y(1) + x(2t) + 2z(3t^2)$. At $t = 1$: $x = y = z = 1$, so $\frac{dw}{dt} = 1 + 2 + 6 = 9$.

3 Let $z = e^{xy}$, $x = t^2$, $y = 1/t$. Find $\frac{dz}{dt}$.
$\frac{\partial z}{\partial x} = ye^{xy}$, $\frac{\partial z}{\partial y} = xe^{xy}$; $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = -t^{-2}$. $\frac{dz}{dt} = ye^{xy}(2t) + xe^{xy}(-t^{-2})$. Since $xy = t^2(1/t) = t$:
$\frac{dz}{dt} = e^t\left(\frac{2t}{t} - \frac{t^2}{t^2}\right) = e^t(2 - 1) = e^t$.

Chain rule with two independent variables

4 Let $z = f(x, y)$ where $x = s + t$, $y = s - t$. Express $\frac{\partial z}{\partial s}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$.

5 For $z = x^2y$, $x = s + t$, $y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $s = 2, t = 1$.
At $s = 2, t = 1$: $x = 3, y = 1$. $\frac{\partial z}{\partial x} = 2xy = 6$, $\frac{\partial z}{\partial y} = x^2 = 9$. Since $\frac{\partial x}{\partial s} = 1$, $\frac{\partial y}{\partial s} = 1$: $\frac{\partial z}{\partial s} = 6 + 9 = 15$. Since $\frac{\partial x}{\partial t} = 1$, $\frac{\partial y}{\partial t} = -1$: $\frac{\partial z}{\partial t} = 6 - 9 = -3$.

6 Let $z = f(x, y)$ where $x = r\cos\theta$, $y = r\sin\theta$ (polar coordinates). Write the chain rule expression for $\dfrac{\partial z}{\partial \theta}$.
$\dfrac{\partial z}{\partial \theta} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial \theta} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial \theta} = \dfrac{\partial z}{\partial x}(-r\sin\theta) + \dfrac{\partial z}{\partial y}(r\cos\theta)$.

Implicit differentiation via the chain rule

7 Use the multivariable chain rule to find $\dfrac{dy}{dx}$ for the implicit curve $x^2y + y^3 = 8$, via the formula $\dfrac{dy}{dx} = -\dfrac{F_x}{F_y}$ where $F(x, y) = x^2y + y^3 - 8$.
$F_x = 2xy$, $F_y = x^2 + 3y^2$. So $\dfrac{dy}{dx} = -\dfrac{2xy}{x^2 + 3y^2}$.

8 For the implicit curve $\sin(xy) = x + y$, find $\dfrac{dy}{dx}$ using $F(x, y) = \sin(xy) - x - y$.
$F_x = y\cos(xy) - 1$, $F_y = x\cos(xy) - 1$. So $\dfrac{dy}{dx} = -\dfrac{y\cos(xy) - 1}{x\cos(xy) - 1} = \dfrac{1 - y\cos(xy)}{x\cos(xy) - 1}$.

Gradient and directional derivative

9 Find the gradient of $f(x, y) = x^2y^3$ at $(1, 2)$.
$\nabla f = \langle 2xy^3, 3x^2y^2\rangle$. At $(1, 2)$: $\langle 2(8), 3(4)\rangle = \langle 16, 12\rangle$.

10 Find the directional derivative of $f(x,y)=x^2y$ at $(2,1)$ in the direction of $\mathbf{v}=\langle 1,1\rangle$.

$\nabla f=\langle 2xy,x^2\rangle$; at $(2,1)$: $\langle 4,4\rangle$. Unit vector $\hat{\mathbf{v}}=\langle\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\rangle$. $D_{\mathbf{v}}f=\frac{4}{\sqrt{2}}+\frac{4}{\sqrt{2}}=\frac{8}{\sqrt{2}}=4\sqrt{2}$.

11 Find the directional derivative of $f(x,y)=e^x\cos y$ at $(0,0)$ in the direction of $\mathbf{v}=\langle 3,4\rangle$.

$\nabla f=\langle e^x\cos y,-e^x\sin y\rangle$; at $(0,0)$: $\langle 1,0\rangle$. Unit vector $\hat{\mathbf{v}}=\langle\frac{3}{5},\frac{4}{5}\rangle$. $D_{\mathbf{v}}f=1\left(\frac{3}{5}\right)+0\left(\frac{4}{5}\right)=\frac{3}{5}$.

The gradient points in the direction of steepest ascent

12 For $f(x,y)=3x^2-y^2$ at $(1,2)$, find the direction of steepest ascent and the maximum rate of increase.

$\nabla f=\langle 6x,-2y\rangle$; at $(1,2)$: $\langle 6,-4\rangle$. Direction of steepest ascent: $\langle 6,-4\rangle$. Maximum rate of increase: $|\nabla f|=\sqrt{36+16}=\sqrt{52}=2\sqrt{13}$.

13 In what direction from $(1,1)$ does $f(x,y)=xy$ decrease most rapidly, and at what rate?

$\nabla f=\langle y,x\rangle$; at $(1,1)$: $\langle 1,1\rangle$. Steepest decrease is $-\nabla f=\langle -1,-1\rangle$, at rate $|\nabla f|=\sqrt{2}$.

14 Explain, using the formula $D_{\mathbf{u}}f=\nabla f\cdot\mathbf{u}=|\nabla f|\cos\theta$ (for unit $\mathbf{u}$), why the direction of steepest ascent is exactly ∇f itself.

Since $|\nabla f|$ is fixed at a given point, $D_{\mathbf{u}}f=|\nabla f|\cos\theta$ is maximized exactly when $\cos\theta=1$, i.e., when $\theta=0$ — meaning $\mathbf{u}$ points in the same direction as ∇f. So among all unit directions, moving along ∇f gives the largest possible directional derivative, equal to $|\nabla f|$.

The gradient is orthogonal to level curves

15 Find ∇f at $(2,1)$ for $f(x,y)=x^2+4y^2$, and state what geometric relationship it has with the level curve $f(x,y)=8$ through that point.

$\nabla f=\langle 2x,8y\rangle$; at $(2,1)$: $\langle 4,8\rangle$. The gradient is always orthogonal (perpendicular) to the level curve passing through that point — here, $\langle 4,8\rangle$ is normal to the ellipse $x^2+4y^2=8$ at $(2,1)$.

16 Find the equation of the tangent line to the level curve $x^2+4y^2=8$ at $(2,1)$, using the fact that $\nabla f=\langle 4,8\rangle$ is normal to it there.

A line with normal vector $\langle 4,8\rangle$ through $(2,1)$ satisfies $4(x-2)+8(y-1)=0$, i.e., $4x+8y=16$, or $x+2y=4$.

Application of the gradient

17 The temperature on a plate is $T(x,y)=100-x^2-2y^2$. A bug at $(2,1)$ wants to cool off fastest. In which direction should it move?

$\nabla T=\langle -2x,-4y\rangle$; at $(2,1)$: $\langle -4,-4\rangle$ (direction of steepest warming). To cool fastest, move in $\langle 4,4\rangle$ (or the unit direction $\langle\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\rangle$).

18 Find the rate of change of $T(x,y) = 100 - x^2 - 2y^2$ at $(2,1)$ in the direction of the unit vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$.

$\nabla T(2,1) = \langle -4, -4 \rangle$ (from above). $D_{\mathbf{v}}T = \langle -4, -4 \rangle \cdot \langle \frac{3}{5}, \frac{4}{5} \rangle = -\frac{12}{5} - \frac{16}{5} = -\frac{28}{5}$.

Related rates via the chain rule

19 The volume of a cylinder is $V = \pi r^2 h$. If r is increasing at 2 cm/s and h is decreasing at 3 cm/s, use the chain rule $\frac{dV}{dt} = \frac{\partial V}{\partial r}\frac{dr}{dt} + \frac{\partial V}{\partial h}\frac{dh}{dt}$ to find $\frac{dV}{dt}$ when $r = 5, h = 10$.

$\frac{\partial V}{\partial r} = 2\pi rh$, $\frac{\partial V}{\partial h} = \pi r^2$. At $r = 5, h = 10$: $\frac{\partial V}{\partial r} = 100\pi$, $\frac{\partial V}{\partial h} = 25\pi$.

$\frac{dV}{dt} = 100\pi(2) + 25\pi(-3) = 200\pi - 75\pi = 125\pi$ cm^3/s.

20 The area of a triangle with sides a, b and included angle θ is $A = \frac{1}{2}ab\sin\theta$. If $a = 5$ cm is fixed, b is increasing at 2 cm/s, and $\theta = \frac{\pi}{6}$ is increasing at 0.1 rad/s, find $\frac{dA}{dt}$ when $b = 8$.

$\frac{\partial A}{\partial b} = \frac{1}{2}a\sin\theta$, $\frac{\partial A}{\partial \theta} = \frac{1}{2}ab\cos\theta$. At $a = 5, b = 8, \theta = \pi/6$: $\frac{\partial A}{\partial b} = \frac{1}{2}(5)(\frac{1}{2}) = \frac{5}{4}$, $\frac{\partial A}{\partial \theta} = \frac{1}{2}(5)(8)\left(\frac{\sqrt{3}}{2}\right) = 10\sqrt{3}$.

So $\frac{dA}{dt} = \frac{5}{4}(2) + 10\sqrt{3}(0.1) = \frac{5}{2} + \sqrt{3} \approx 4.23$ cm^2/s.

The gradient in three dimensions

21 Find ∇f for $f(x,y,z) = x^2y + yz^2$ at the point $(1,2,3)$.

$\nabla f = \langle 2xy,\ x^2 + z^2,\ 2yz \rangle$. At $(1,2,3)$: $\langle 4,\ 10,\ 12 \rangle$.

22 Find the directional derivative of $f(x,y,z) = xyz$ at $(1,1,1)$ in the direction of $\mathbf{v} = \langle 1,1,1 \rangle$.

$\nabla f = \langle yz, xz, xy \rangle$; at $(1,1,1)$: $\langle 1,1,1 \rangle$. Unit vector $\hat{\mathbf{v}} = \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$. $D_{\mathbf{v}}f = \frac{1+1+1}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$.

23 Find the direction of steepest ascent of $f(x,y,z) = x^2 + y^2 + z^2$ at $(1,2,2)$, and the maximum rate of increase.

$\nabla f = \langle 2x, 2y, 2z \rangle$; at $(1,2,2)$: $\langle 2,4,4 \rangle$. This is the direction of steepest ascent, with maximum rate $|\nabla f| = \sqrt{4 + 16 + 16} = \sqrt{36} = 6$.

24 Explain briefly why ∇f for $f(x,y,z) = x^2 + y^2 + z^2$ always points radially outward from the origin.

$\nabla f = \langle 2x, 2y, 2z \rangle = 2\langle x,y,z \rangle$ is just twice the position vector, which always points directly away from the origin — so the level surfaces (spheres centered at the origin) grow fastest exactly in the outward radial direction, matching the geometric intuition that moving straight away from the center increases $x^2 + y^2 + z^2$ fastest.

25 For $f(x,y,z) = xy + yz + zx$, find ∇f at $(1,-1,2)$ and the directional derivative in the direction of $\mathbf{v} = \langle 0,0,1 \rangle$.

$\nabla f = \langle y+z, x+z, y+x \rangle$; at $(1,-1,2)$: $\langle 1,3,0 \rangle$. Since $\mathbf{v} = \langle 0,0,1 \rangle$ is already a unit vector, $D_{\mathbf{v}}f = \langle 1,3,0 \rangle \cdot \langle 0,0,1 \rangle = 0$.